Lesson 1 Definitions, Theorems, and Proofs

In Math 1365, we learn to do *proofs*, to show if a math statement is true (or false), using the language of *rigorous* mathematics based on *logic*. In logic,

- A **statement** is a declarative sentence that has a definite **truth value**, that is, it is **true** or **false** but not both simultaneously.

Determine if each of the following is a statement. If it is, is it true or false?

a) **The sum of two even numbers is even.** It is a true statement.
b) $\mathbf{1 + 1 = 3}$ It is a false statement.
c) $\boldsymbol{x + 3 = 5}$ It is a statement, true when $x = 2$ and false when $x \neq 2$.
d) $\boldsymbol{x + y}$ It is not a statement; it does *not* declare and has *no truth value*.
e) **This sentence is false.** It is not a statement as it is self-contradictory with *no definite truth value*: if it is true, then it is false, if it is false, then it is true.
f) **What is Goldbach's conjecture**? It is a question, *not* a declarative statement.

Only a statement, which has a unique *truth value* (true or false), can be proved or disproved. A non-statement, such as (d), (e), or (f), cannot be proved or disproved.

This course will revolve around three types of *mathematical* statements:

- A **definition** gives precise meaning to a term, and it is always true!
- A **theorem** is a statement that has been proved to be true.
- A **proof** is a set of statements organized logically to show that a theorem is true.

We give our first example of definitions, theorems, and proofs.

Definition An integer x is ***even*** if $x = 2n$ for some integer n.
Theorem 8 is an even integer.
Proof Since $8 = 2 \cdot 4$, and 4 is an integer, 8 is even by definition of even integers.

A proved statement is a **theorem**. A more modest word for theorem is **proposition** or a **result**. We use "**fact**" for any true statement, be it a definition or a theorem.

- In a proof, *logic* is employed to make *new* conclusions from *given* information.

Our simple example introduces *the process of rigorous proof*. There are **3 *given* facts**: i) definition of even integers, ii) $8 = 2 \cdot 4$, and iii) 4 is an integer, from which we get **a *new* conclusion**: 8 is even, using ***logic***, although the logic is so simple and natural in this case that we hardly notice it!

Writing proof is an *art*; we have to decide *what* to include. Do we mention the definitions or "obvious" facts like "4 is an integer"? As beginners, we might include more, but in time, we may want a *minimal* proof: $8 = 2 \cdot 4$. At any stage, it is important that we *know clearly* what we are doing. Before we continue, we give a list of **undefined terms** whose meanings *everyone knows* and **unproven statements** (**axioms**) whose truth *everyone accepts*.

Numbers and operations. We assume that everyone knows a **real number** and an **integer**, as well as **positive** and **negative** numbers, and that **0** is neither positive nor negative. But we will define even and odd integers, rationals, and other numbers. We know how to **add**, **subtract**, **multiply**, and **divide** numbers. We know without proof that the sum or product of integers (reals) is an integer (real), but we will prove that the sum of two even numbers is even.

Properties of arithmetic. We use the following laws (*without* explicit mention):

- Addition and multiplication are **commutative**: $x + y = y + x,\ \ xy = yx$
- $+$ and $\times$ are **associative**: $(x + y) + z = x + (y + z),\ (xy)z = x(yz)$
- The **distributive** law: $x(y + z) = xy + xz$

Equality and ordering of real numbers. For example, $3 > 1$, $-2.5 < 1$, etc.

There are *many* ways to define a thing. We may define "even" using *division*:

An integer x is ***even*** if $x/2$ is an integer.

(Previously we used *multiplication* $x = 2n$.) The proof for "8 is even" is then:

Since $8/2 = 4$ is an integer, 8 is even.

To prove that a statement is false is to ***disprove*** it. We disprove that 3 is even, in two ways, depending on which definitions we use:

a) 3 is not even because $3 \neq 2n$ for any integer n.
b) 3 is not even because $3/2 = 1.5$ is not an integer.

Unless otherwise stated, we use the definitions marked Definition in the lessons.

We move on to define **odd** integers and a fundamental concept: **divisibility**.

Definition An integer x is ***odd*** if $x = 2n + 1$ for some integer n.
Definition An integer a ***divides*** an integer b, denoted $a|b$, if $b = an$ for some integer n; a is a ***divisor/factor*** of b, and b is a ***multiple*** of a.
Definition An integer $p > 1$ is ***prime*** if the only positive divisors of p are 1 and p.

Pay attention to the notation $a|b$, it is *not* the same as a/b! We use $a \nmid b$ to denote "a does *not* divide b"; *slash* is used to denote "not". Note that *evenness* is a special case of divisibility: "even" means "divisible by 2".

We prove the following propositions using the corresponding definitions.

a) 9 is odd, since $9 = 2(4) + 1$.
b) $4|12$, since $12 = 4(3)$.
c) 5 is prime, since $5 > 1$ and the only positive divisors of 5 are 1 and 5.
d) 15 is not prime: Since $15 = 3(5)$, 15 has a positive divisor 3, which is neither 1 nor 15.

Some proofs may feel just like *explanations*. Yes, a proof *is* an explanation, but rigorous and based on *definitions*. There are many ways to write a good (or bad) proof. With practice and attention, we can develop a taste for good proofs and the skills to write one.